

Design Thinking Unit 4: Innovation

SECTION A

Very Short Questions

1. Define innovation.

Innovation is the process of converting creative ideas into practical, useful, and value-adding solutions.

2. What is creativity?

Creativity refers to imagining or generating new and original ideas.

3. Differentiate invention and innovation in one line.

Invention is creating something new; innovation is applying it to create value.

4. What is product innovation?

It is the creation or improvement of tangible products.

5. What is process innovation?

Process innovation improves how a product is made or how a service is delivered.

6. Give one example of business model innovation.

Example: Netflix shifting from DVD rentals to online streaming.

7. What does TRL stand for?

TRL stands for Technology Readiness Level.

8. What does TRL 1 represent?

Basic principles observed; initial idea stage.

9. What does TRL 9 represent?

Technology fully deployed in real-world operational environment.

10. What is marketing innovation?

It means adopting new ways of promoting, pricing, or positioning products.

11. Give an example of organizational innovation.

Google's "20% time" policy.

12. Define grassroots innovation.

Innovations developed by common people to solve local problems using simple resources.

13. What is social innovation?

Solutions created to improve society and benefit communities.

14. Name one Indian grassroots innovator.

Mansukhbhai Prajapati (Mitticool fridge).

15. What is meant by value creation?

Delivering benefits that customers are willing to pay for or adopt.

16. Write one benefit of innovation.

Innovation improves the effectiveness and efficiency of solutions.

17. Give an example of process innovation in banking.

UPI mobile payments.

18. Define business model.

A business model explains how a company creates, delivers, and captures value.

19. What is incremental innovation?

Small improvements made to existing products or processes.

20. What is radical innovation?

A major breakthrough that changes industries or creates new ones.

Short Questions

1. Distinguish between product and process innovation.

Product innovation changes the product itself by improving features or creating new ones.
Process innovation changes how the product is produced or delivered to improve efficiency, cost, or speed.

2. Explain the role of creativity in innovation.

Creativity generates original ideas, while innovation turns those ideas into useful applications.
Without creativity, innovation has no starting point.

3. What is business model innovation? Give an example.

Changing how a business earns money or delivers value.
Example: Uber enabling ride-hailing without owning cars.

4. Explain marketing innovation with an example.

Marketing innovation uses new promotional or pricing techniques.
Example: Coca-Cola's "Share a Coke" name-printing campaign.

5. What are the benefits of process innovation?

- Reduces cost
- Improves speed
- Enhances service quality
- Increases reliability

6. Give two examples of social innovation.

- Community kitchens for underprivileged groups
- Low-cost sanitary pads for rural women

7. Explain TRL levels in brief.

TRLs measure technology maturity from idea (TRL 1) to full deployment (TRL 9).

8. What is incremental innovation? Give an example.

Small continuous improvements.

Example: Adding more megapixels to mobile cameras every year.

9. What is radical innovation? Give an example.

Disruptive new solutions.

Example: Introduction of smartphones replacing basic phones.

10. Mention two characteristics of grassroots innovation.

- Low-cost solutions
- Designed by local people for local problems

11. What is user-driven innovation?

Innovation based on observing or obtaining insights from users' behaviors, needs, and challenges.

12. Explain why innovation is important for businesses.

It helps companies stay competitive, meet customer needs, improve efficiency, and create new revenue streams.

Detailed Questions

1. Explain with examples the four major types of innovation: product, process, business model, and marketing.

A1.a.Product Innovation

Product innovation involves creating new products or significantly improving existing ones to better meet user needs or solve problems more effectively. It focuses on enhancing features, performance, usability, and value. A classic example is smartphones replacing basic mobile phones—combining

calling, internet access, cameras, and apps into one powerful, modern device that transformed user expectations and daily life.

b.Process Innovation

Process innovation improves how a product is manufactured or how a service is delivered, aiming to increase efficiency, reduce cost, and enhance speed and convenience. It often uses technology or smarter workflow design. Online banking and UPI payments are strong examples—reducing queues, enabling instant money transfer, minimizing paperwork, and making financial services accessible anytime, anywhere.

c.Business Model Innovation

Business model innovation changes how a company creates, delivers, or captures value. It rethinks revenue methods, customer access, or value delivery systems. Netflix demonstrated this by moving from DVD rentals to online streaming—shifting from physical distribution to a digital, subscription-based model. This not only changed how audiences consume content but also reshaped the entire entertainment industry.

d.Marketing Innovation

Marketing innovation introduces new methods to promote, position, price, or sell products, aiming to attract and engage customers more effectively. It goes beyond traditional advertising and uses creative, data-driven, or technology-based approaches. Examples include personalized ads based on user interests and influencer marketing, where trusted creators recommend products—both increasing reach, relevance, and customer trust.

2. Explain grassroots innovation with two real Indian examples.

A2.Grassroots innovation emerges from ordinary people who directly experience everyday problems and design practical, low-cost solutions using locally available materials. These innovations are driven not by profit but by necessity, creativity, and a deep understanding of real-life challenges within their communities. They are simple, environmentally friendly, and socially meaningful. A well-known example is the **Mitticool Fridge** by Mansukhbhai Prajapati—a clay refrigerator that works without electricity, making it ideal for rural households with unreliable power. Another impactful example is the **Jaipur Foot**, a low-cost, durable prosthetic leg that has restored mobility and dignity to thousands of economically disadvantaged individuals. Grassroots innovation shows how powerful solutions can arise from modest resources and genuine human need.

3. Describe social innovation and differentiate it from business innovation.

Social innovation refers to solutions created to improve societal well-being by addressing issues like inequality, health, education, and community development. Its focus is on welfare, inclusivity, social change, and long-term human impact rather than profit. In contrast, **business innovation** aims to strengthen a company's competitiveness through improved products, services, or models that enhance revenue, customer satisfaction, and operational efficiency. While both involve creativity and problem-solving, their intentions differ: social innovation prioritizes social impact and collective benefit, whereas business innovation prioritizes value creation, growth, and competitive advantage. Together, they show how innovation can serve both society and industry in different ways.

Basis	Social Innovation	Business Innovation
Primary Goal	Improve societal well-being and solve community problems	Increase profits, competitiveness, and business growth
Focus Areas	Welfare, inclusivity, social change, public benefit	Customers, revenue, efficiency, market advantage
Outcome Priority	Social impact and long-term community improvement	Value creation, customer satisfaction, financial returns
Beneficiaries	Society at large, vulnerable groups, communities	Business, customers, stakeholders
Driving Force	Need-based problem solving for social good	Market demand, competition, profitability

4. What is the importance of TRL levels? Explain TRL 1, TRL 5, TRL 9.

A4.TRL helps assess how mature a technology is before adopting or investing.

- **TRL 1:** Basic principle observed; theoretical idea.
- **TRL 5:** Technology validated in relevant environment; prototype testing.
- **TRL 9:** Actual technology proven and deployed in real operations.

5. Identify the types of innovation Uber introduced and explain each.

- a. **Business Model Innovation:** Users book rides through an app; Uber owns no cars.
- b. **Process Innovation:** Real-time GPS tracking, fare estimation, digital payments.
- c. **Marketing Innovation:** Referral bonuses, dynamic pricing strategies.
- d. **Organizational Innovation:** Gig-based driver model.

Q6. How did Amul combine process and organizational innovation?

A6. Amul’s success comes from blending **process innovation** with **organizational innovation** in a way that transformed India’s dairy sector. On the process side, Amul introduced automated milk collection centers equipped with fat-testing machines, cold storage systems, and streamlined logistics that ensured fair pricing, reduced spoilage, and improved overall quality. These technological steps made milk collection faster, transparent, and more efficient. On the organizational side, Amul adopted the cooperative model, enabling millions of small farmers to become stakeholders rather than suppliers. This structure empowered rural communities, ensured stable incomes, and created a highly coordinated, democratic supply chain that still thrives today

Q7. Explain why UPI adoption is considered process innovation.

A7. UPI adoption is considered **process innovation** because it fundamentally transformed the way financial transactions are carried out in India. Instead of relying on cash, cheques, or lengthy bank procedures, UPI introduced an instant, seamless, and paperless method for transferring money using mobile phones. It simplified authentication, eliminated delays, reduced dependency on physical bank visits, and significantly cut transaction costs. For banks, UPI streamlined back-end operations, reduced workload at branches, and enhanced customer service efficiency. For users, it improved convenience, accessibility, and financial inclusion. This re-engineering of the transaction process is exactly what defines process innovation.

Q8.Identify two innovations behind TikTok’s success.

A8.Product Innovation: Easy video editing tools, filters, and effects.

Marketing Innovation: Algorithm-driven content makes ordinary users go viral, attracting millions.

9. Explain radical vs incremental innovation with suitable examples.

A9.

Basis	Radical Innovation	Incremental Innovation
Definition	Major breakthroughs that create entirely new technologies, markets, or ways of doing things.	Small, continuous improvements made to existing products, services, or processes.
Impact Level	High impact; disrupts industries and changes consumer behaviour.	Low to moderate impact; enhances user experience gradually.
Risk Factor	High risk due to uncertainty and high investment.	Low risk because it builds on existing systems.
Speed of Change	Sudden, transformative change.	Slow, progressive change over time.
Resources Required	Requires substantial R&D, new skills, and major technological shifts.	Uses existing knowledge, infrastructure, and capabilities.
Examples	Smartphones replacing feature phones; electric vehicles replacing petrol cars; digital photography replacing film cameras.	Improving camera megapixels in phones; adding new flavors to packaged foods; software updates in apps.

Q10. Why is e-Choupal considered social & business innovation?

A. e-Choupal, initiated by ITC, serves both social and business purposes. Socially, it empowers rural farmers by providing direct access to market information, weather updates, and best farming practices, improving income, productivity, and livelihoods. Farmers gain knowledge and agency, fostering community development. From a business perspective, ITC reduces transaction costs, ensures better quality raw materials, and streamlines its supply chain, enhancing competitiveness and profitability. By integrating technology with rural needs, e-Choupal exemplifies an innovation that simultaneously drives social welfare and business efficiency, making it a classic example of **dual-purpose innovation** in emerging economies.